Introduction to Linear Programming

Julia Wrobel

Overview

Today, we cover:

- Talk about final project proposals
- Linear programming

Announcements

- Sign up for final project proposals slot

Final project presentations

Sign up for final project presentation slot!

- On Canvas, go to calendar
- Do this now.

Final project presentations

Presentation is designed to be in the style of a JSM or ENAR contributed talk

- Presentation should be 15 minutes or less
- Aim for 12-13 minutes to leave time for questions
- Practice! You will be deducted for not finishing in the alloted time

Final project presentations

Students are required to provide brief but meaningful feedback on all presentations. There will be online forms provided.

- You will receive feedback from your peers anonymously

Introduction to LP

Linear programming (LP) or **linear optimization** is a set of (constrained) optimization algorithms.

- began in 1947 when George Dantzig devised the simplex method
- LP and **quadratic programming** are two special cases of convex optimization with linear constraints

Widely used across industries!

Linear programming

An optimization problem with a linear objective function and linear constraint functions is called a linear program (LP):

$$\begin{aligned} \text{minimize } & c^T x + d \\ \text{subject to } & Ax \preceq b \end{aligned}$$

- **Recall**: you can always convert a minimization problem to a maximization problem by multiplying the objective function by -1
- In statistics, applications include LASSO, quantile regression, support vector machine

Linear programming techniques

Methods for solving LP problems:

- Graphical method
- Simplex algorithm
- Interior point methods

Linear programming techniques

rq {quantreg} R Documentation

Quantile Regression

Description

Returns an object of class "rq" "rqs" or "rq.process" that represents a quantile regression fit.

`method` the algorithmic method used to compute the fit. There are several options:

1. "br" The default method is the modified version of the Barrodale and Roberts algorithm for l_1-regression, used by `l1fit` in S, and is described in detail in Koenker and d'Orey(1987, 1994), default = "br". This is quite efficient for problems up to several thousand observations, and may be used to compute the full quantile regression process. It also implements a scheme for computing confidence intervals for the estimated parameters, based on inversion of a rank test described in Koenker(1994).
2. "fn" For larger problems it is advantageous to use the Frisch–Newton interior point method "fn". This is described in detail in Portnoy and Koenker(1997).
3. "pfn" For even larger problems one can use the Frisch–Newton approach

Carpenter problem

A carpenter can make tables or she can make bookshelves.

- **Table**: costs 10 units of lumber and 5 hours of labor
  - Provides $180 in profit
- **Bookcase**: costs 20 units of lumber, 4 hours of labor
  - $200 in profit
- Only 200 units of lumber available
- 80 hours of labor available

What number of tables and bookcases provide the optimal profit?

Carpenter problem

- **Table**: costs 10 units of lumber and 5 hours of labor, $180 in profit
- **Bookcase**: costs 20 units of lumber, 4 hours of labor, $200 in profit
- 200 units of lumber available; 80 hours of labor available

Let x_1 and x_2 denote number of tables, bookshelves respectively

- $10x_1 + 20x_2 \leq 200$
- $5x_1 + 4x_2 \leq 80$
- $x_1, x_2 \geq 0$
- maximize $f(x_1, x_2) = 180x_1 + 200x_2$

Carpenter problem

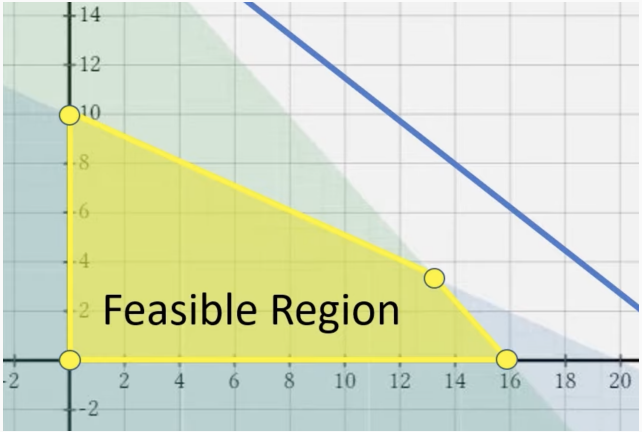
Solving LP graphically

The small 2-variable problem can be represented in a 2D graph.

- An **isoprofit line** connects all combinations of x_1 and x_2 that yield the same profit value
  - e.g., $180x_1 + 200x_2 = k$ for different values of k
- The isoprofit line is perpendicular to the gradient of the objective function
- **Key idea**: slide the isoprofit line outward (increasing k) until it just touches the feasible region
  - that contact point is optimal

Feasible region

Shaded region that satisfies constraints is called feasible region



Properties of the optimal solution

Optimal solution might not exist

- the objective function is unbounded in the solution space (can be improved indefinitely without violating any constraints)
- the feasible region is empty (no solution satisfies all constraints)

But if the optimal solutions exists:

- Interior point: No.
- Corner point: Yes. The corner points are often referred to as "extreme points" or vertices.
- Edge point: Yes, only when the edge is parallel to the isoprofit line. In this case, every point on that edge has the same objective function value, including the corner point of that edge.
- **Key result**: If an LP has an optimal solution, it must have a extreme point optimal solution. This greatly reduces our search of optimal solutions: only need to check a finite number of extreme points.

Big ideas of LP

- Each inequality constraint defines a half-plane; their intersection is the **feasible region**
- The feasible region is convex (no "dents")
- Extreme values (min and max) occur at the **vertices** of this region
- For 2 variables: graphical method works. For p variables: need an algorithm.

You can solve this algorithmically (rather than graphically) using the **simplex method**.

Simplex method

Linear programming involves optimizing (maximizing or minimizing) a linear objective function, subject to a set of linear constraints. The Simplex method helps to find the optimal solution to such problems, provided one exists.

For simplex method:

- the LP problem needs to be in **standard** form
  - objective function set up as a maximization problem
  - constraints set up as linear inequalities of the form $Ax \leq b$
  - need all variables x to be nonnegative

LP Standard form

An LP can be expressed using only equality and nonnegativity constraints. This is called **standard form**:

$$\begin{aligned} \text{minimize } & c^T x \\ \text{subject to } & Ax = b \\ & x \geq 0 \end{aligned}$$

An LP can also be expressed using only inequality constraints. This is called **inequality form**:

$$\begin{aligned} \text{minimize } & c^T x \\ \text{subject to } & Ax \leq b \end{aligned}$$

Slack variables

Are added to less than or equal to inequality constraints to transform them into equality constraints

- represent the "unused" capacity within a constraint
- non-negative variables
- Say you have inequality $2x + 3y \leq 10$.
  - Convert to to equality constraint $2x + 3y + s = 10$
  - If $s \neq 0$, then there the combination of x and y could be increased further without violating the constraint

Simplex: convert to standard form using slack variables

Go back to Carpenter example.

- maximize $z = f(x_1, x_2) = 180x_1 + 200x_2$
- $x_1, x_2 \geq 0$

Constraints:

- $10x_1 + 20x_2 \leq 200 \rightarrow 10x_1 + 20x_2 + s_1 = 200$
- $5x_1 + 4x_2 \leq 80 \rightarrow 5x_1 + 4x_2 + s_2 = 80$

Slack variables pick up any "slack" in the system based on current values of x_1 and x_2

- $s_1, s_2 \geq 0$
- $s_1, s_2 = 0$ when z is maximized

Steps of Simplex algorithm

1. Convert problem to standard form
2. Set up "tableau": basically amounts to writing out problem as a matrix
3. Start at a vertex of the feasible region, then pivot

- Essentially, you are iteratively moving from one vertex (or "corner point") of the feasible region to another, with the goal of improving the objective function at each step.

Initial tableau

First $z = 180x_1 + 200x_2 \rightarrow z - 180x_1 - 200x_2 = 0$

- $10x_1 + 20x_2 \leq 200 \rightarrow 10x_1 + 20x_2 + s_1 = 200$
- $5x_1 + 4x_2 \leq 80 \rightarrow 5x_1 + 4x_2 + s_2 = 80$

Start at $x_1, x_2 = 0$

Pivot step

Choose the entering variable (which x_j to increase from zero?):

- Look at the objective row: $z - 180x_1 - 200x_2 = 0$
- The most negative coefficient determines the entering variable
- Here: x_2 enters (coefficient -200 is more negative than -180)

Pivot step

Choose the leaving variable (minimum ratio test):

- Divide each RHS by the entering variable's coefficient in that row
- s_1: $200/20 = 10$; s_2: $80/4 = 20$
- The row with the **smallest non-negative ratio** leaves: s_1 leaves
- This keeps all variables non-negative (stays in the feasible region)

Pivot step

Perform row operations to make the pivot column a unit vector:

- Divide the pivot row by the pivot element (20)
- Eliminate x_2 from all other rows using row operations
- Update z row and the RHS
- Repeat until no negative coefficients remain in the z row

Solution

Summary of the Simplex method

Basic variables define current corner point/vertex!

- Those are the variables that are literally a unit basis of the matrix
- Walks along the edge of the feasible space to improve the objective function
  - "Basic" variables are the current solution
- Only considers vertices
- Moves as far as it can in optimal direction until movement is blocked by another constraint

Standard form as a matrix

$$\begin{aligned} \max z &= c^T x \\ s.t. Ax &= b \\ x &\geq 0 \end{aligned}$$

$$x = [x_1, x_2, s_1, s_2]^T$$

- Extending x to include slack variables s_1, s_2 converts all inequalities to equalities
- The full augmented vector is what the simplex algorithm operates on

Linear programming techniques

- Graphical method
  - Only works for 2 variables
- Simplex algorithm
  - Moves from one basic feasible solution to the next
  - Standard version works for $\leq$ constraints
- Two phase, big M methods
  - Variations on Simplex method
  - Handle $=$ and $\geq$ constraints
- Interior point methods
  - Better for large-scale LP problems
  - Instead of moving along boundary, moves through interior of feasible region

Interior point methods

Effectively, the idea is to set up the problem such that we can solve it using something like Newton's method.

- Need to set up some background and theory first
- Allows us to handle the constraints

Carpenter problem

A carpenter can make tables or she can make bookshelves.

- **Table**: costs 10 units of lumber and 5 hours of labor
  - Provides $180 in profit
- **Bookcase**: costs 20 units of lumber, 4 hours of labor
  - $200 in profit
- Only 200 units of lumber available
- 80 hours of labor available

What number of tables and bookcases provide the optimal profit?

LP problem for resource allocation

Take a typical maximizing LP problem (3 variables and 2 constraints).

$$
\begin{aligned}
\max \quad & c_1 x_1 + c_2 x_2 + c_3 x_3 \\
\text{s.t.} \quad & a_{11} x_1 + a_{12} x_2 + a_{13} x_3 \leq b_1 \\
& a_{21} x_1 + a_{22} x_2 + a_{23} x_3 \leq b_2 \\
& x_1, x_2, x_3 \geq 0
\end{aligned}
$$

Economic interpretation of the problem:

- We produce three products using two materials
- x_j: unit of production of product $j \in \{1, 2, 3\}$. Unknown to obtain.
- c_j: profit per unit of product
- a_{ij}: unit of material ($i \in \{1, 2\}$) to produce 1 unit of product j
- b_i: units of available material i.

Goal is to maximize the profit, subject to the material constraints.

Resource valuation problem

Now, suppose a buyer is considering purchasing our entire inventory of materials but is unsure how to price them. The buyer knows we will only agree to the sale if it yields a higher return than using the materials for production.

Buyer's business strategy: producing one fewer unit of product j will save us:

- a_{1j} units of material 1 and a_{2j} units of material 2

The buyer seeks to determine the unit prices of materials to minimize costs while ensuring we still agree to the sale (i.e., we do not earn less). Let y_1 and y_2 be the unknown unit prices of the materials. The buyer's optimization problem, known as the **Resource Valuation Problem**, is then:

Resource valuation problem

$$
\begin{aligned}
\max \quad & c_1x_1 + c_2x_2 + c_3x_3 \\
\text{s.t.} \quad & a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \leq b_1 \\
& a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \leq b_2 \\
& x_1, x_2, x_3 \geq 0
\end{aligned}
$$

$$
\begin{aligned}
\min \quad & b_1y_1 + b_2y_2 \\
\text{s.t.} \quad & a_{11}y_1 + a_{21}y_2 \geq c_1 \\
& a_{12}y_1 + a_{22}y_2 \geq c_2 \\
& a_{13}y_1 + a_{23}y_2 \geq c_3 \\
& y_1, y_2 \geq 0
\end{aligned}
$$

Duality

The buyer's LP problem is called the "**dual**" problem of the original problem, which is called the "**primal** problem.

In matrix notation, if the **primal** LP problem is:

$$\begin{aligned} \max \quad & cx \\ \text{s.t.} \quad & Ax \leq b, x \geq 0 \end{aligned}$$

The corresponding **dual** problem is:

$$\begin{aligned} \min \quad & b^T y \\ \text{s.t.} \quad & A^T y \geq c^T, y \geq 0 \end{aligned}$$

Duality

We can also express the dual problem in canonical form (a maximization problem with $\leq$ constraints):

$$\begin{aligned} \max \quad & -b^T y \\ \text{s.t.} \quad & -A^T y \leq -c^T, y \geq 0 \end{aligned}$$

The **dual is the negative transpose of the primal**.

- i.e. the dual of the dual problem is the primal problem

Why do we care about duality?

Duality

Why do we care about duality?

- Sometimes the dual problem is easier to solve than the primal one
- If a constraint is slightly relaxed or tightened, the dual variables indicate how much the objective function will change
- Solution to primal is **always** $\leq$ solution to dual (weak duality theorem)!

Need to understand conditions in which the dual and primal solutions are equal!

Duality in non-canonical form

What if the primal problem doesn't fit into the canonical form (e.g., has $\geq$ or $=$ constraints, unrestricted variables)?

The **variable types in the dual problem** are determined by the **constraint types in the primal problem** as follows:

- Equality constraints in the primal $\rightarrow$ Unrestricted dual variables
- $\leq$ in the primal $\rightarrow$ **nonnegative** dual variables
- $\geq$ in the primal $\rightarrow$ **nonpositive** dual variables

i.e.

Primal (max) constraints	**Dual (min) variable**
$\leq$	≥ 0
$\geq$	≤ 0
$=$	unrestricted

Duality in non-canonical form

Conversely, **constraint types of the dual** problem are determined by **variable types of the primal**:

Primal (max) variable	**Dual (min) constraints**
≥ 0	$\geq$
≤ 0	$\leq$
unrestricted	$=$

Non-canonical duality example

If the primal problem is:

$$
\begin{aligned}
max \quad & 20x_1 + 10x_2 + 50x_3 \\
s.t. \quad & 3x_1 + x_2 + 9x_3 \leq 10 \\
& 7x_1 + 2x_2 + 3x_3 = 8 \\
& 6x_1 + x_2 + 10x_3 \geq 1 \\
& x_1 \geq 0, \quad x_2 \textit{ unrestricted}, \quad x_3 \leq 0
\end{aligned}
$$

The dual problem is:

Non-canonical duality example

If the primal problem is:

$$
\begin{aligned}
max \quad & 20x_1 + 10x_2 + 50x_3 \\
s.t. \quad & 3x_1 + x_2 + 9x_3 \leq 10 \\
& 7x_1 + 2x_2 + 3x_3 = 8 \\
& 6x_1 + x_2 + 10x_3 \geq 1 \\
& x_1 \geq 0, \quad x_2 \textit{ unrestricted}, \quad x_3 \leq 0
\end{aligned}
$$

The dual problem is:

$$
\begin{aligned}
min \quad & 10y_1 + 8y_2 + y_3 \\
s.t. \quad & 3y_1 + 7y_2 + 6y_3 \geq 20 \\
& y_1 + 2y_2 + y_3 = 10 \\
& 9y_1 + 3y_2 + 10y_3 \leq 50 \\
& y_1 \geq 0, \quad y_2 \textit{ unrestricted}, \quad y_3 \leq 0
\end{aligned}
$$

Weak duality

Weak duality theorem:

- For any feasible solution x to the **primal** (maximization) problem and any feasible solution y to the **dual** (minimization) problem:

$$\text{objective of dual} \;\geq\; \text{objective of primal.}$$

- Implications:
  - The dual provides a bound on the primal's optimal value.
  - Useful for proving infeasibility or bounds in optimization problems.
- **Weak Duality Guarantee**: the optimal value of the primal cannot exceed the optimal value of the dual.

In other words, we will only do business if selling the material makes us more money.

Weak duality proof

Weak duality: If $(x_1, \dots, x_n)$ is a feasible solution for the primal, and $(y_1, \dots, y_m)$ is a feasible solution for the dual, then $\sum_j c_j x_j \leq \sum_i b_i y_i$.

Weak duality proof

Weak duality: If $(x_1, \dots, x_n)$ is a feasible solution for the primal, and $(y_1, \dots, y_m)$ is a feasible solution for the dual, then $\sum_j c_j x_j \leq \sum_i b_i y_i$.

$$
\begin{aligned}
\sum_j c_j x_j &\leq \sum_j \left(\sum_i y_i a_{ij} \right) x_j \\
&= \sum_{i,j} y_i a_{ij} x_j \\
&= \sum_i \left(\sum_j a_{ij} x_j \right) y_i \\
&\leq \sum_i b_i y_i .
\end{aligned}
$$

Strong duality

- We know that at feasible solutions for both dual and primal, the solution to the dual is always greater or equal
- If there is a difference between the largest primal value and the smallest dual value it is called the "**Duality gap**".
    - When are these values equal?

Strong Duality Theorem: If the primal has an optimal solution, then the dual also has an optimal solution and there is no duality gap, i.e.:

$$\text{objective of dual} \;=\; \text{objective of primal.}$$

- Condition for strong duality: the feasible region must be nonempty and bounded.
- The **duality gap** is zero when strong duality holds.

Complementary slackness theorem

The **complementary slackness theorem** provides a necessary condition for optimality in linear programming. It states that:

If x^* and y^* are feasible solutions of primal and dual problems, then x^* and y^* are both optimal if and only if

1. $y^{*T}(b - Ax^*) = 0$
2. $(y^{*T}A - c)x^* = 0$

This theorem is the foundation of the **interior point method** class of LP solvers.

Complementary slackness theorem

Consider a simple LP problem:

$$\begin{aligned} \textit{max} \quad & z = x_1 + x_2, \\ \textit{s.t.} \quad & x_1 + 2x_2 \leq 100, \\ & 2x_1 + x_2 \leq 100, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Its dual problem is:

Complementary slackness theorem

Consider a simple LP problem:

$$\begin{aligned} \textit{max} \quad & z = x_1 + x_2, \\ \textit{s.t.} \quad & x_1 + 2x_2 \leq 100, \\ & 2x_1 + x_2 \leq 100, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Its dual problem is:

$$\begin{aligned} \textit{min} \quad & z = 100y_1 + 100y_2, \\ \textit{s.t.} \quad & y_1 + 2y_2 \geq 1, \\ & 2y_1 + y_2 \geq 1, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Complementary slackness theorem

Complementary slackness theorem

The complementary slackness theorem states:

$$
\begin{aligned}
y_1(100 - x_1 - 2x_2) &= 0, \\
y_2(100 - 2x_1 - x_2) &= 0, \\
x_1(y_1 + 2y_2 - 1) &= 0, \\
x_2(2y_1 + y_2 - 1) &= 0.
\end{aligned}
$$

At the optimal point, we have $\mathbf{x} = (\mathbf{100/3}, \mathbf{100/3})$, $\mathbf{y} = (\mathbf{1/3}, \mathbf{1/3})$. The complementary slackness holds.

- All the constraints are bounded in both primal and dual problems.

Plot of constraints

Complementary slackness theorem

Modify the simple LP problem a bit:

$$\begin{aligned} \textit{max} \quad & z = 3x_1 + x_2, \\ \textit{s.t.} \quad & x_1 + 2x_2 \leq 100, \\ & 2x_1 + x_2 \leq 100, \\ & x_1, x_2 \geq 0. \end{aligned}$$

and its dual problem is:

$$\begin{aligned} \textit{min} \quad & z = 100y_1 + 100y_2, \\ \textit{s.t.} \quad & y_1 + 2y_2 \geq 3, \\ & 2y_1 + y_2 \geq 1, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Complementary slackness theorem

Now the complementary slackness theorem states:

$$
\begin{aligned}
y_1(100 - x_1 - 2x_2) &= 0, \\
y_2(100 - 2x_1 - x_2) &= 0, \\
x_1(y_1 + 2y_2 - 3) &= 0, \\
x_2(2y_1 + y_2 - 1) &= 0.
\end{aligned}
$$

For this LP problem, the optimal solutions are $\mathbf{x} = (\mathbf{50}, \mathbf{0})$, $\mathbf{y} = (\mathbf{0}, \mathbf{1.5})$. Complementary slackness still holds. We observe that:

- In the primal problem:
  - The first constraint is **inactive** (slack > 0) at the optimum, so its shadow price $y_1 = 0$
  - Second constraint is **active** (binding, slack $= 0$), so y_2 can be non-zero

Complementary slackness theorem

For this LP problem, the optimal solutions are $\mathbf{x} = (\mathbf{50}, \mathbf{0})$, $\mathbf{y} = (\mathbf{0}, \mathbf{1.5})$. Complementary slackness still holds. We observe that:

- In the primal problem:
  - The first constraint is **inactive** (slack > 0), so its shadow price $y_1 = 0$.
  - The second constraint is **active** (binding, slack $= 0$), so y_2 can be non-zero.
- In the dual problem:
  - The first constraint is **active** (binding), so its primal variable x_1 can be non-zero.
  - The second constraint is **inactive** (slack > 0), so its primal variable $x_2 = 0$.

Plot of constraints

Economics interpretation

Dual variables can be called "**shadow prices**" of the primal constraint, how objective function would increase if constraint was relaxed.

- If a primal constraint is bounded, relaxing that constraint would result in a gain (improve the objective function), shadow price is non-zero.
- If a primal constraint is unbounded, relaxing that constraint would not improve the objective function, shadow price is zero.

Economics interpretation

Bakery example: Imagine you're running a bakery, and can make a limited number of cakes because you have a limited amount of flour. Shadow price tells you how much more profit you'd gain if you had a little more flour.

- If you're tight on flour, getting more flour would let you bake more cakes and make more money. - In this case, the shadow price is nonzero because the extra resource improves your outcome.
- If you have plenty of flour, adding more doesn't help—you're already making as many cakes as you can sell. In this case, the shadow price is zero because extra flour doesn't change your profit.

So, the dual variable (shadow price) tells you how valuable it would be to relax a constraint—meaning allowing more of that constrained resource. If relaxing the constraint helps, the shadow price is positive. If it doesn't make a difference, the shadow price is zero.

Primal/dual optimality conditions

Given the primal and dual problem with slack/surplus variables added:

Primal:

$$\begin{aligned} &\textit{max} \quad cx \\ &\textit{s.t.} \quad Ax + w = b, \quad x, w \geq 0. \end{aligned}$$

Dual:

$$\begin{aligned} &\textit{min} \quad b^T y \\ &\textit{s.t.} \quad A^T y - z = c^T, \quad y, z \geq 0. \end{aligned}$$

The *complementary slackness theorem* states that at optimal solution, we should have $x_j z_j = 0 \forall j$, and $w_i y_i = 0 \forall i$. In matrix notation, the complementary conditions can be rewritten as

$$XZe = 0, WYe = 0$$

- X, Z, W, Y are diagonal matrices
- e is a vector of 1's

Primal/dual optimality conditions

We now have the optimality conditions for the primal/dual problems as:

$$
\begin{aligned}
Ax + w - b &= 0, \\
A^T y - z - c^T &= 0, \\
XZe &= 0, \\
WYe &= 0, \\
x, y, w, z &\geq 0.
\end{aligned}
$$

- The first two conditions are simply the constraints for primal/dual problems
- The next two are complementary slackness
- The last one is the non-negativity constraint

Primal/dual optimality conditions

We now have the optimality conditions for the primal/dual problems as:

$$
\begin{aligned}
Ax + w - b &= 0, \\
A^T y - z - c^T &= 0, \\
XZe &= 0, \\
WYe &= 0, \\
x, y, w, z &\geq 0.
\end{aligned}
$$

Ignoring the non-negativity constraints, this is a set of $2n + 2m$ equations with $2n + 2m$ unknowns (n and m are the number of unknowns and constraints in the primal problem), which can be solved using Newton's method.

Primal-dual interior point method

The primal-dual interior point method finds the primal-dual optimal solution (x^*, y^*, w^*, z^*) by applying Newton's method to the primal-dual optimality conditions.

- The direction and length of the steps are modified in each step so that the non-negativity condition is strictly satisfied in each iteration.

To be specific, define the following function $F : R^{2n+2m} \to R^{2n+2m}$:

$$\mathbf{F}(x, y, w, z) = \begin{bmatrix} \mathbf{A}x + \mathbf{w} - \mathbf{b} \\ \mathbf{A}^T y - \mathbf{z} - \mathbf{c}^T \\ XZe \\ WYe \end{bmatrix}$$

The goal is to find solution for $F = 0$.

Primal-dual interior point method

Applying Newton's method, if at iteration k the variables are (x^k, y^k, w^k, z^k), we obtain a search direction
$(\delta x, \delta y, \delta w, \delta z)$ by solving the linear equations:

$$F'(x^k, y^k, w^k, z^k) \begin{bmatrix} \delta x \\ \delta y \\ \delta w \\ \delta z \end{bmatrix} = -F(x^k, y^k, w^k, z^k).$$

Primal-dual interior point method

Then the update will be:

$$(x^{k+1}, y^{k+1}, w^{k+1}, z^{k+1}) = (x^k, y^k, w^k, z^k) + \alpha(\delta x, \delta y, \delta w, \delta z)$$

with $\alpha \in (0, 1]$ chosen so that the result from the next iteration is feasible.

- Solving for $(\delta x, \delta y, \delta w, \delta z)$ using the equation on the previous slide enforces feasibility of solutions

An improved algorithm

The algorithm in its current setup is not ideal because often the steps are small to avoid violating the positivity constraints. Instead:

- The value of $XZe + WYe$ represents the duality gap
- Instead of trying to eliminate the duality gap (set it to 0), reduce the duality gap by some factor in each step

We replace the complementary slackness by:

$$XZe = \mu_x e \tag{1}$$
$$WYe = \mu_y e \tag{2}$$

When $\mu_x, \mu_y \to 0$ as $k \to \infty$ the solution from this system will converge to the optimal solution of the original LP problem. Good selections of μ are $\mu_x^{k+1} = (x^k)^T z^k / n$ and $\mu_y^{k+1} = (w^k)^T y^k / m$

- n and m are the dimensions of x and y

An improved algorithm

Under the new algorithm, at the kth iteration, the Newton equations become:

$$\begin{bmatrix} A & 0 & I & -0 \\ 0 & A^T & 0 & -I \\ Z & 0 & 0 & X \\ 0 & W & Y & 0 \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \\ \delta w \\ \delta z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -X^k Z^k e + \mu_x^k e \\ -W^k Y^k e + \mu_y^k e \end{bmatrix}$$

This provides the **general primal-dual interior point method** as follows:

1. Choose strictly feasible initial solution (x^0, y^0, w^0, z^0), and set $k = 0$. Then repeat (2) and (3) until convergence:
2. Solve system (1) to obtain the updates $(\delta x, \delta y, \delta w, \delta z)$
3. Update the solution $(x^{k+1}, y^{k+1}, w^{k+1}, z^{k+1}) = (x^k, y^k, w^k, z^k) + \alpha^k(\delta x, \delta y, \delta w, \delta z)$. α^k is chosen so that all variables are ≥ 0.

Lagrange multipliers

The method of Lagrange multiplier is a general algorithm for optimization problems with equality constraints. For example, consider a problem:

$$\begin{aligned} \max \quad & f(x, y) \\ s.t. \quad & g(x, y) = c \end{aligned}$$

We introduce a new variable λ called a **Lagrange multiplier** and form the following new objective function:

$$L(x, y, \lambda) = f(x, y) + \lambda[g(x, y) - c]$$

We will then optimize L with respect to x, y and λ using typical methods. Note that the condition $\partial L / \partial \lambda = 0$ at the optimal solution guarantees that the constraints will be satisfied.

The barrier problem

The interior point algorithm is closely related to the **Barrier Problem**. Go back to the primal problem:

$$\begin{aligned} \max \quad & z = cx \\ s.t. \quad & Ax + w = b; x, w \geq 0 \end{aligned}$$

The non-negativity constraints can be replaced by adding two **barrier terms** in the objective function. The barrier term is defined as $B(x) = \sum_j \log x_j$, which is finite as long as x_j is positive. Then the primal problem becomes:

$$\begin{aligned} \max \quad & z = cx + \mu_x B(x) + \mu_y B(w) \\ s.t. \quad & Ax + w = b \end{aligned}$$

The barrier terms make sure x and w won't become negative.

The barrier problem

The Lagrangian for this problem is (using y as the multiplier):

$$L(x, y, w) = cx + \mu_x B(x) + \mu_y B(w) + y^T(b - w - Ax)$$

The optimal solution for the problem satisfies:

$$c + \mu_x X^{-1} e - A^T y = 0 \tag{3}$$
$$b - w - Ax = 0 \tag{4}$$
$$\mu_y W^{-1} e - y = 0 \tag{5}$$

Define new variables $z = \mu_x X^{-1} e$ and rewrite these conditions, we obtain exactly the same set of equations as the relaxed optimality conditions for primal-dual problem. **Note**: $z = \mu_x X^{-1} e$ is also a condition so there are 4 in total.